

Was the Universe an Accident?

Artificial intelligence may find the answer in data from the Large Hadron Collider

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How I got from your chair to here

- Born in England. The first ten years of my life in Iran. Landed in California at ten
- I had some English, from a summer school the year before. They put me in ESL anyway. I hated it and got myself out as fast as I could
- Irvine, California. Honors and AP classes
- Junior year: calculus, and Physics C
- Senior year: driving to UC Irvine for linear algebra, differential equations, vector calculus

I was sitting exactly where you are, in 1992.

What I was actually like

- My favorite toy, at two or three, was a screwdriver
- I took everything apart to see how it worked
- At ten that became computers, and I started programming
- RC cars. Models. Real programming projects at twelve and thirteen
- A programming course at a community college, at that age

That is the whole story. Everything after this is the same instinct with a bigger budget.

- Physics major. Grueling — I will not pretend otherwise
- What was worth having was the room: everyone around me was smart, and I learned as much from the people next to me as from the faculty
- **I did research from my first year** — the Artificial Intelligence lab first, then physics
- Undergraduate thesis on an experiment at Fermilab

Undergraduates do real research. Mostly it is a matter of asking.

Then it was a long road

- PhD at the University of Maryland, on an experiment at the Stanford Linear Accelerator
- Four years living next to the beamline in California
- Then Chicago
- Then the Large Hadron Collider, which is where I have been since

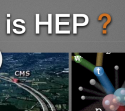
What I actually do

What is HEP ?



Large Hadron Collider (LHC)

Largest Machine Ever Built



Create New Particles
Only 1 in 10^{11} are "interesting"



Record with 5 Story
100M Channel
"Camera"
(60 TB/s)



10^{11} Protons Collide 40 Million Times per Second



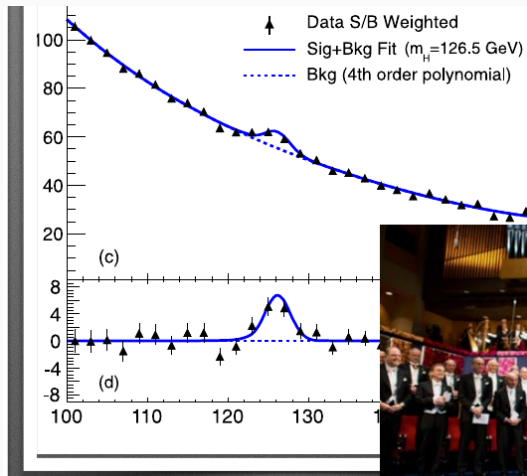
Processed by
1M Cores
Around the World

There is no looking at this data

- 10^{11} protons in a bunch, colliding **40 million times a second**
- A detector that is a **100-million-channel camera**, reading out at **60 terabytes per second**
- Processed by **about a million cores** around the world
- **One collision in 10^{11}** is interesting

Nobody looks at this data. There is no looking. Code is the only way in.

What we found, in 2012



What that plot is

- 2012: a new particle. The last missing piece of the Standard Model — the line on this shirt
- Every term on it is explained, slide by slide, in the repo at the end of this talk
- Nobel Prize in Physics, 2013
- About 6,000 people on the experiment. I was one of them

That plot is how we announced a new particle. Look at what it is: a histogram. You will make histograms this year.

Why artificial intelligence

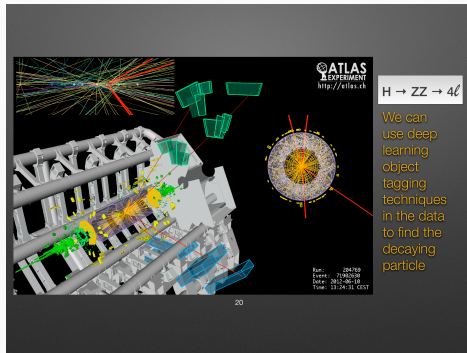
We are not done

- The Higgs came out **too light**. The theory only works if a number is tuned to absurd precision
- So either the universe got lucky — an accident, or a multiverse where we happen to live in the one that works
- Or there is a mechanism we have not found yet: supersymmetry, extra dimensions, substructure

This is the actual open question. It is why I still have a job.

Needles in 10^{11} haystacks

- Nobody knows how to write down "interesting" as a rule
- So you show a machine a few million examples and let it learn the difference
- And it has to decide **while the beam is still running** — on data we will never store



Not a chatbot. A decision about what to keep, made in nanoseconds, 40 million times a second.

Why I ended up doing data science

The lever

- I started programming at ten, so by the time I got to physics, the computing was the part I was already good at
- In a field where the data is too big to look at, **the person who can make the computer look is the person who gets to answer the question**
- That kept compounding: better computing, better analysis, harder questions — and eventually AI, which is what I now lead in my field

The physics was always the goal. The computing was the lever — and the lever kept getting more valuable.

My field had this problem first

- **The Web was invented at CERN**, by a physicist, to share physics data
- Worldwide computing grids — about a million cores today — and petabytes a year, since the 1990s
- Machine learning to identify particles, also since the 1990s, decades before it was fashionable
- In 2014, colleagues of mine put a real ATLAS problem on Kaggle as a public competition

We were data scientists before the job existed. The data forced it.

Why science and data science

Take my own field. It is almost ideally suited:

- Physics systems and machine learning systems hit the **same problems**
- **Decades of experience** at the data frontier
- A bridge between science and industry
- Physicists are engineers by training — we build the instrument too

MOVE OVER, CODERS— PHYSICISTS WILL SOON RULE SILICON VALLEY

... it's happening across Silicon Valley. ..., *the things that just about every internet company needs to do are more and more suited to the skill set of a physicist.*

new wave of data science and AI is something that suits physicists right down to their socks.

"There is something very natural about a physicist going into machine learning ... more natural than a computer scientist."

Physicists know how to handle data ... building these enormously complex systems requires its own breed of abstract thought.

Every scientist is becoming one

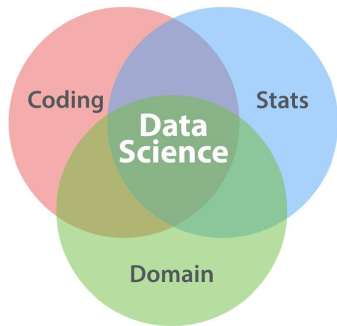
- Telescopes, genome sequencers, mass spectrometers, climate models — every field now produces more data than any person can look at
- The scarce skill is not the algorithm. It is **someone who understands the measurement and can also write the code**
- A data scientist who does not know your field cannot tell a real effect from an artifact of your instrument. You can

Scientists make the best data scientists. The science is the part that takes years.

It takes three things

- **Coding**
- **Math**, especially statistics
- **Domain expertise** — chemistry, physics, biology, whatever you actually care about

Each one is a degree on its own. Most people arrive with one and pick up the others late.



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Which is exactly why our degree is in the College of Science — not in computer science, and not in a business school.

The door is already open to you

The degree

- A **BS in Data Science**, in the College of Science at UT Arlington
- Start from zero as a freshman — no preparation assumed
- Pick a concentration: **chemistry**, physics, biology, math, psychology, earth and environmental science
- Ends in a capstone project you choose

uta.edu/science/data-science

You do not have to wait

- I took college classes while I was still in high school. You can take ours
- **A colleague's son took this exact course as a high school senior, ten years ago**

The door is not something that opens after graduation. It is open now.

Take this home

Everything for the course I am teaching right now
— public, free, no account needed:

- `Chapter.01.Computers.pdf` — how computers work
- `Chapter.03...Statistics.pdf` — uncertainty, distributions, error bars
- `Lecture.13.Slides.pdf` — how to look at data. Useful for your project this year
- `Lecture.9.Standard-Model-Shirt.pdf` — **this shirt**, line by line
- `Labs/Lab.1` — how to set a laptop up to do any of this



[github.com/UTA-DataScience/
DATA3402.Fall.2026](https://github.com/UTA-DataScience/DATA3402.Fall.2026)

The part nobody tells you

- A science PhD is **funded**. Tuition paid, and you are paid a stipend
- You teach first, then you are a research assistant doing your own research
- Graduate students and postdocs do most of the actual work in science. You are not a student in the back of the room — you are the workforce
- After the PhD comes a postdoc, and then, if you are lucky, a faculty job. That is a steep down-select, and it is not the only path. Most people from my field go to industry — into exactly these data science and AI jobs

You can get paid to get a PhD. Almost nobody tells you that in high school.

Your project

Four things, whatever your project is on

1. **Look at your data before you model it.** Histogram everything you measured, first — before any fit, before any conclusion. I have 10^{11} collisions and cannot look at mine. You have two hundred trials. You can look at yours
2. **Every count carries $\sqrt{n}$.** Twelve events is 12 ± 3.5 . Without error bars you cannot yet say two conditions differ
3. **A null result is a result** — if you can say which kind. “No effect” and “not enough data to see an effect” are different claims, and knowing which one you have is the most impressive thing you can show a judge
4. **Never show a plot you have not read yourself.** Bringing me a plot you know is broken is fine. Bringing one that is broken without noticing is not

The same question

The question that got me from a chair in Irvine to a hole in the ground in Switzerland is the same one you are asking this year:

How does this actually work?

Take the thing apart. Look at your data. The rest is budget.

github.com/UTA-DataScience/DATA3402.Fall.2026

uta.edu/science/data-science